Anim. Behav., 1986, **34**, 1182–1188

Discrimination between full-sibling spiny mice (*Acomys cahirinus*) by olfactory signatures

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Abstract. Discrimination between individual full siblings was tested in spiny mice (*Acomys cahirinus*). At weaning, intact litters were separated into two pairs so that animals only had social contact with one of their full siblings. Following 10 days of being housed in pairs, the four weanlings from a given litter were reunited in an observation terrarium. Pairings by familiar siblings were more frequent than unfamiliar-sibling pairings, indicating that interactions among close kin are mediated by the recognition of individual signatures rather than general phenotypic traits shared by all members of a kin class. Animals rendered anosmic by intranasal injections of zinc sulphate displayed no evidence of discriminating between familiar and unfamiliar siblings. Olfactory cues therefore appear to function as the signatures by which individual siblings are recognized. Since all animals were fed identical diets and otherwise maintained in similar environments, the differences in olfactory signatures between full siblings are most likely a function of genetic variability within this class of kin.

Detailed studies of animal social behaviour have revealed numerous examples of social recognition (see Colgan 1983 for a recent review). The term individual recognition implies that particular animals can be identified or discriminated from other conspecifics. Similarly, the related concept of kin recognition refers to an ability to discriminate between kin and non-kin. Kin recognition, at least in its broad sense, can therefore be inferred from discriminative interactions among close kin as compared to unrelated, or only distantly related, conspecifics. As with any form of social recognition, kin recognition is dependent upon phenotypic traits or signatures that establish an individual's identity or membership in a particular class or group (Alexander 1979; Beecher 1982). While the mechanisms mediating discrimination of kin signatures have been investigated for numerous species (e.g. Blaustein 1983; Holmes & Sherman 1983), the characteristics of these salient signatures have typically not been well elucidated.

It is theoretically possible that recognition of kin could involve discrimination of particular conspecifics by their individually unique signatures. At the opposite extreme, members of a given kin class (for example, littermate full siblings) could share similar signatures and accordingly be treated indiscriminately (e.g. Waldman 1981). In the latter case, kin signatures would be truly distinct from individual signatures.

Previous research in our laboratory suggests that full-sibling spiny mice (*Acomys cahirinus*) may share somewhat similar phenotypic signatures that facilitate littermate recognition. Littermates who had been separated for 9 days and then reunited displayed evidence of mutual recognition, providing that they had been housed with another member of their same litter during the separation period (Porter et al. 1983). In contrast, animals isolated for 8 days subsequently displayed no signs of littermate recognition (Porter & Wyrick 1979). Based upon these data, one may infer that spiny mice learn the phenotypic traits of conspecifics with which they are housed and are thereby able to recognize age-mates whose signatures presumably resemble that of the familiar conspecifics. Thus, signatures of familiar kin may be used as standards against which to compare the signatures of unfamiliar animals.

The degree of overlap in the signatures of close *A. cahirinus* kin (full siblings) and the sensory basis of such phenotypic traits are investigated further in the present series of experiments. The major question being addressed is whether the signatures of littermate full siblings are functionally identical or, alternatively, sufficiently dissimilar to allow for the discrimination between individuals within this kin class.

Related studies with laboratory rats (Wills et al. 1983), mice (Kareem 1983) and ground squirrels

(Holmes 1984) have found that familiar siblings were discriminated from unfamiliar siblings that had been raised with a foster mother and littermates. Such manipulations allow one to directly assess the role of familiarity in kin recognition, but the basis of the signatures mediating discrimination between familiar and unfamiliar kin following differential fostering remains ambiguous. Siblings reared apart could be discriminated through phenotypic differences resulting from underlying genotypic variability or by distinctive labels (e.g. maternal labels or colony-specific odours) acquired from the foster conspecifics with which they had been housed (Brown 1979; Porter et al. 1981, 1984).

In an attempt to prevent this problem, in the following experiments sibling familiarity was manipulated by separating intact litters into pairs at weaning, so that animals only had social contact with one of their full siblings from that time until the beginning of the recognition tests. Therefore, since all weanlings from a given litter were otherwise treated identically, any discrimination between familiar and unfamiliar littermates would presumably entail genetically-based variability between their signatures, even if these signatures are acquired from social partners. That is, the production of individually recognizable labels transferred among the full siblings would itself be a reflection of genetic differences between these close kin.

EXPERIMENT I

Methods

Litters, comprised of four to five spiny mouse (*Acomys cahirinus*) full siblings, were randomly selected from the laboratory breeding colony. Family units containing both parents and their newborn litter were maintained in individual home cages until the beginning of the experimental treatment on days 18–22 postpartum. At that time, four littermates were removed from their home-cage, marked for the purpose of individual identification and housed as pairs in two separate cages measuring 18 × 29 × 13 cm high. These two littermate pairs remained separated from one another for a period of 10 full days and were maintained on ad libitum water and Wayne mouse breeder chow (the same diet fed to the breeding colony). The two littermates housed together during the 10-day treatment period will henceforth be referred to as familiar siblings and those siblings housed apart as unfamiliar.

Immediately following the 10-day pair housing treatment, tests for discrimination between familiar and unfamiliar littermate siblings were conducted. At the start of the 5-day test session, the four littermates that had been housed in two separate pairs were placed together in a glass terrarium measuring 29 × 60 × 30 cm high, with the floor covered by a shallow layer of bedding material (sugar-cane waste). These four siblings were then observed during six 5-min observation sessions per day for 5 consecutive days; the first observations were conducted at least 60 min after the animals were initially placed into the observation cage. At the beginning of each 5-min observation session and at the end of each 60-s interval therein, the identities of all animals that were physically touching one another (i.e. touching of limbs, head, or trunk; excluding touching of whiskers or tails) were recorded. Accordingly, there was a total of six observation points for each 5-min session. The six 5-min observation sessions were distributed throughout the day with at least 30 min between successive sessions.

In this manner, 10 groups (each containing four littermate full siblings) were observed during the light phase of the 13:11 h light/dark cycle. Each of these 10 groups was composed of animals of either sex (there was at least one male and one female in each four-animal group).

Results

As for previous research on kin interactions among spiny mice, social recognition can be inferred if pairings by particular animals occur more frequently than expected by chance alone. Observed frequencies of pairings by familiar and unfamiliar littermates are presented in Table I. Over the 5-day observation period, the maximum number of possible pairings would be 360 for each group of four animals. Since pairings by unfamiliar littermates could arise from twice as many of the possible combinations of two animals in each observation cage as could familiar-littermate pairings, the expected ratio of unfamiliar to familiar pairings was 2:1 with each group. To correcct for this expected 2:1 ratio, the observed frequency of unfamiliar-sibling pairings was multiplied by 0·75 and the frequency of pairings by familiar siblings

Table I. Observed frequencies of pairings by familiar and unfamiliar littermate full siblings (in mixed-sex groups) over the 5-day test period

	Familiar pairs	Unfamiliar pairs
Uncorrected median	209·5	1·2
Uncorrected mean	201·1	0·5
	(range = 0–319)	(range = 0–3)
Corrected median*	314·3	0·9

* Results of Wilcoxon's matched-pairs signed-ranks test comparing corrected medians of familiar versus unfamiliar pairings: $Z = 2{\cdot}67$, N (minus ties) $= 9$, $P < 0{\cdot}01$ (two-tailed).

Table II. Observed frequencies of pairings by familiar and unfamiliar littermate full siblings (in single-sexed groups) over the 5-day test period

	Familiar pairs	Unfamiliar pairs
Uncorrected median	127·5	0·5
Uncorrected mean	108·1	3·9
	(range = 2–222)	(range = 0–19)
Corrected median*	191·3	0·4

* Results of Wilcoxon's matched-pairs signed-ranks test comparing corrected medians of familiar versus unfamiliar pairings: $Z = 2{\cdot}80$, $N = 10$, $P < 0{\cdot}005$ (two-tailed).

by 1·5. The resulting corrected frequencies of pairing by familiar versus unfamiliar siblings were compared with Wilcoxon's matched-pairs signed-ranks test. As seen in Table I, pairings by familiar littermates were significantly more frequent than were pairings by unfamiliar littermates.

EXPERIMENT II

In the previous experiment, each group of four siblings contained animals of both sexes. Therefore, it is possible that discrimination between familiar and unfamiliar littermates could have been mediated at least partially by sex-specific traits rather than individual signatures alone. To rule out the potential influence of sex-specific phenotypic traits on discrimination between individual siblings, the above experiment was replicated with sibling groups each comprised of animals of a single sex.

Methods

Ten groups, each containing four littermate full siblings of a single sex, were observed in the second experiment. Five of these groups were each made up of four male siblings and the remaining five groups each contained four full-sister littermates. These 10 groups of siblings were obtained from separate litters born in the laboratory breeding colony and were removed from their respective home-cages at 20–23 days after birth and housed in littermate pairs. Recognition tests began following 10 days of the pair-housing treatment. Further procedural details are otherwise identical to those described above for the previous experiment.

Results

Frequencies of pairings by familiar and unfamiliar littermates of the same sex are presented in Table II. Statistical comparisons (Wilcoxon's test) of the corrected frequencies indicate that pairs of familiar siblings were observed significantly more than were pairings by unfamiliar siblings. In each of the 10 cages, the (corrected) frequency of pairings by familiar siblings was greater than that for unfamiliar littermates. Thus, both the male and the female groups consistently displayed preferential familiar-sibling pairings.

EXPERIMENT III

The results of earlier studies with *A. cahirinus* indicate that discrimination between familiar siblings and unfamiliar unrelated age-mates is based upon olfactory cues (Porter et al. 1978). Animals rendered anosmic through intranasal injection of zinc sulphate displayed no evidence of recognizing familiar littermate siblings. Holmes (1984) has recently reported comparable data following zinc sulphate treatment in thirteen-lined ground squirrels.

The final experiment in the present series attempted to ascertain whether odours are likewise implicated in the discrimination between familiar and unfamiliar *A. cahirinus* siblings. In other words, do olfactory cues function as salient signatures mediating recognition of individual littermate full siblings?

Methods

Twenty mixed-sex groups, each containing four littermate full siblings, were removed from their respective home-cages on day 18–24 after birth. At that time, each four-animal group was divided into two pairs and each of these sibling pairs was housed in an individual cage for a period of 10 days. As in the above described experiments, siblings housed together during the 10 days of pair housing will be designated familiar, those housed apart, unfamiliar siblings. At the end of this 10-day period, sibling groups (four littermates that had been housed as two separate pairs) were randomly assigned to the following two treatment conditions.

Olfactory impairment

Ten groups of siblings were included in the olfactory impairment condition. Upon removal from their pair-housing cages, all four animals in each group were treated with zinc sulphate, a substance known to induce temporary anosmia in a variety of rodents, including *A. cahirinus* (Alberts & Galef 1971; Porter et al. 1978; Holmes 1984). Each animal was lightly anaesthetized with ether and 0·02–0·03 ml of 5% zinc sulphate solution was injected into each nostril through a blunt 26-gauge needle inserted 1–2 mm into the nostril. As soon as the animals appeared to have recovered from the effects of the ether, the two pairs of separated littermates composing a sibling group were placed together into an observation cage where all instances of physical contact were recorded during 5-min observation sessions over 5 consecutive days. Further details of the observation procedures and pair-housing treatment are otherwise identical to those of the preceding experiments.

Following the last observation session on day 5, the olfactory capabilities of the animals in each group were assessed by burying two pieces of fresh apple under the bedding material and recording the latency (with a maximum of 15 min) until either piece of apple was uncovered by any animal in the cage. Since this method of assessing olfactory impairment involves the simultaneous testing of all four animals in a group, it is not as rigorous as comparable tests involving one animal at a time. On the other hand, such simultaneous testing entails minimal disruption or handling of the animals prior to the test.

Distilled water control

In the control condition, 10 groups of four siblings (each of mixed-sex) were treated and tested in the same manner as the animals in the olfactory impairment condition, with a single exception; viz. 0·02–0·03 ml of distilled water, rather than zinc sulphate, was injected into each nostril. At the end of the 5-day observation period, control animals were also tested for olfactory sensitivity by recording their latencies to find a piece of buried apple.

Results

Corrected and uncorrected frequencies of pairings by familiar and unfamiliar siblings are presented separately in Table III for the olfactory impairment (zinc sulphate) and control (distilled water) conditions. Animals rendered anosmic with zinc sulphate did not differ in their frequencies of pairings by familiar versus unfamiliar siblings. In

Table III. Observed frequencies of pairings by familiar and unfamiliar full siblings in the olfactory impairment and control conditions

	Olfactory impairment (zinc sulphate) condition		Control (distilled water) condition	
	Familiar pairs	Unfamiliar pairs	Familiar pairs	Unfamiliar pairs
Uncorrected median	0·8	14·0	98·5	0·8
Uncorrected mean	3·1	23·1	124·4	2·5
Range	0–12	0–79	0–298	0–21
Corrected median*	1·2	10·5	147·8	0·6

* Results of Wilcoxon's matched-pairs signed-ranks tests comparing corrected medians of familiar versus unfamiliar pairings: for the olfactory impairment condition, $Z = 1{\cdot}60$, N (minus ties) $= 9$, NS; for the distilled water control condition, $Z = 2{\cdot}67$, $N = 10$, $P < 0{\cdot}01$ (two-tailed).

Table IV. Observed frequencies of pairings by familiar siblings and huddling by all four animals in an observation cage for the olfactory impairment and control conditions

	Familiar pairings		Four-animal huddles	
	Zinc sulphate condition	Distilled water condition	Zinc sulphate condition	Distilled water condition
Mean	3·1	124·4	94·1	2·4
Range	0–12	0–298	0–141	0–12
Median*	0·8	98·5	110·0	7·3

* Results of Mann–Whitney *U*-tests comparing medians of the zinc sulphate versus control conditions: for familiar pairings, $Z=3{\cdot}32$, $P<0{\cdot}001$ (two-tailed); for the four-animal huddles, $Z=3{\cdot}49$, $P<0{\cdot}0005$ (two-tailed).

contrast, the distilled water control animals were observed in familiar pairs significantly more frequently than in unfamiliar pairs.

Additional statistical analyses directly compared the data for animals across the two treatment conditions (see Table IV). The observed (uncorrected) frequencies of pairing by familiar siblings was significantly greater among animals treated with distilled water than with zinc sulphate. Rather than remaining in pairs, animals treated with zinc sulphate tended to huddle in a heterogeneous group containing all four animals (familiar and unfamiliar) in a cage. Such four-animal huddles were significantly more frequent in the olfactory impairment condition than in the control condition.

Finally, the analyses of the buried apple test following the last observation session reveal that animals treated with zinc sulphate took significantly longer than the distilled water controls to locate the apple ($Z=3{\cdot}86$, Mann–Whitney *U*-test; $P<0{\cdot}001$). Mean latencies until finding the apple were 757·6 s (SD = 259·4) for the olfactory impairment condition and 94·1 s (SD = 20·0) for the distilled water control condition.

DISCUSSION

As is evident from the results of the first two experiments, *A. cahirinus* weanlings can discriminate between individual full siblings from their own litter, even when these littermates are all of the same sex. Preferential huddling among familiar as opposed to unfamiliar full siblings implies that sibling interactions are based primarily upon the recognition of individual signatures rather than on a general signature shared by all members of a kin class (or by individuals of the same sex and age within a kin class) (cf. Hölldobler & Michener 1980; Breed & Bekoff 1981). Individuals become familiar with the signatures of conspecifics to which they are exposed and therefore are able to distinguish between those animals and unfamiliar conspecifics, including other close relatives.

In contrast with the data presented above, previous studies indicate that there may be at least some perceptible similarity in the signatures of full-sibling *A. cahirinus* (Porter et al. 1983). Weanlings housed in littermate pairs subsequently recognized other members of their same litter, suggesting that the signatures of sibling cage-mates were used as a standard against which to compare the phenotypes of others. Animals housed without such sibling contact after weaning showed no signs of sibling recognition. Thus, although phenotype-matching may play some role in kin recognition, this mechanism appears to be of limited importance and may only become salient when recognition through the process of direct contact and familiarization is precluded. Any influence of phenotype-matching as a mechanism mediating sibling recognition in the present series of experiments was completely masked by the effects of familiarity with particular individuals (see Lacy & Sherman 1983, for a further discussion of phenotype-matching).

Since zinc-sulphate-treated animals responded indiscriminately to familiar and unfamiliar siblings, whereas the distilled water controls displayed preferential familiar-sibling pairings, it can be concluded that olfactory signatures mediate the recognition of individual siblings. Results of the buried-apple test provide further evidence of olfactory impairment among those animals treated with zinc sulphate. Five days after the intranasal injections of zinc sulphate, the olfactory sensitivity of animals in that condition still appeared to be deficient relative to that of the control animals. Olfaction has now been implicated as the primary modality involved in various forms of social recognition in *A. cahirinus*, including parental recognition of offspring (Doane & Porter 1978; Makin & Porter 1984); pups' responsiveness to lactating females (Porter & Doane 1976); discrimination between littermates and unfamiliar age-mates (Porter et al. 1978) and, in the present instance, discrimination between individual littermates.

Discernible differences in the olfactory signatures of full siblings are most probably a function of genetic variability within this kin class. Because environmental variables, such as dietary constituents, can also influence individual odours (e.g. Leon 1975), familiar and unfamiliar siblings in the present study were maintained in similar environments which included identical diets. It is therefore unlikely that differential environmental influences could account for the differences in olfactory phenotypes between these siblings.

An ability to distinguish between individual close kin would allow for a more complex social system tham would be possible if animals were simply recognized as members or non-members of a particular kin class, but not otherwise differentiated. Recognition of individual kin could result in the formation of subgroups (alliances), differential treatment of individuals, and social ranks or hierarchies, rather than indiscriminate interactions amongst members of a homogeneous group. Animals who could discriminate between kin might benefit (in a genetic sense) by allocating a disproportionate share of their nepotistic behaviour or resources to those relatives that are most likely to survive and reproduce, or to those with whom reciprocal sharing of resources is practised. As pointed out by Rothstein (1980), reciprocal altruism might occur amongst kin as well as unrelated individuals and would be dependent upon individual recognition in each instance.

ACKNOWLEDGMENTS

This research project was supported by grant number HD15051 from the National Institute of Child Health and Human Development.

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(Received 10 June 1985; revised 5 August 1985; MS. number: A4557)